THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The exponent of 3 in the prime factorisation of 2250 is: 1

(a) 5 **(b)** 2 **(c)** 3 **(d)** 4

2. The number of zeroes of a polynomial whose graph cuts the x-axis at 1 point(s) is: 1

(a) 2 **(b)** 0 **(c)** 3 **(d)** 1

3. The pair of equations $3x + 2y = 5$ and $9x + 6y = 15$ is: 1

(a) exactly two solutions
(b) no solution (parallel lines)
(c) infinitely many solutions (coincident lines)
(d) exactly one solution (intersecting lines)

4. For the equation $x^2 - 7x + 12 = 0$, the sum of the roots is: 1

(a) -7 **(b)** 7 **(c)** 12 **(d)** -5

5. The 10th term of $21, 18, 15, 12, \ldots$ is: 1

(a) -3 **(b)** -6 **(c)** -9 **(d)** -30

6. In $\triangle ABC$, $DE \parallel BC$, $AD = 1$ cm, $DB = 3$ cm, $AE = 2$ cm. Then EC (in cm) $=$ 1

(a) 2 **(b)** 3 **(c)** 6 **(d)** 7

7. The distance between $(2, 3)$ and $(5, 7)$ is: 1

(a) 25 **(b)** 5 **(c)** 1 **(d)** 7

8. If $\tan A = \frac{4}{3}$, then $\sin A =$ 1

(a) $\frac{3}{5}$ **(b)** $\frac{4}{5}$ **(c)** $\frac{5}{4}$ **(d)** $\frac{4}{3}$

9. From a point on level ground, 15 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) $30\sqrt{3}$ m **(b)** $5\sqrt{3}$ m **(c)** $15\sqrt{3}$ m **(d)** 15 m

10. If tangents PA and PB from an external point P to a circle with centre O are inclined to each other at 80°, then $\angle POA$ is: 1

(a) 40° **(b)** 100° **(c)** 80° **(d)** 50°

11. A circular field has radius 14 m. Fencing costs ₹ 10 per metre. The cost (in ₹) of fencing the field once is: 1

(a) ₹ 880 **(b)** ₹ 88 **(c)** ₹ 440 **(d)** ₹ 6160

12. A toy is a cone of base radius 7 cm and height 24 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 858 cm² **(b)** 550 cm² **(c)** 1012 cm² **(d)** 836 cm²

13. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 3, 16, 20, 4, 7 respectively, The mean of the data is: 1

(a) 442 **(b)** 45 **(c)** 45.2 **(d)** 44.2

14. Two dice are thrown simultaneously. The probability that the sum of the numbers is 8 is: 1

(a) $\frac{5}{36}$ **(b)** $\frac{31}{36}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{1}{9}$

15. If HCF(306, 657) = 9, then LCM(306, 657) is: 1

(a) 201042 **(b)** 22338 **(c)** 963 **(d)** 2754

16. If one zero of $x^2 - 7x + 10$ is 2, the other zero is: 1

(a) 9 **(b)** -5 **(c)** 10 **(d)** 5

17. The solution of $x + y = 8$, $x - y = 2$ has $x =$ 1

(a) 8 **(b)** 5 **(c)** 3 **(d)** -5

18. For $x^2 - 6x + k = 0$ to have equal roots, $k =$ 1

(a) 6 **(b)** 3 **(c)** 9 **(d)** 36

19. The 5th term from the end of the AP with last term 100 and $d = 4$ is: 1

(a) 116 **(b)** 84 **(c)** 80 **(d)** 88

20. $\triangle ABC \sim \triangle DEF$ with areas 16 cm² and 36 cm². If $AB = 6$ cm, then DE (cm) $=$ 1

(a) 6 **(b)** 9 **(c)** 27 **(d)** 12

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Check whether 6^n can end with the digit 0 for any natural number n. 2

22. The graph of $y = p(x)$ cuts the x-axis at $x = -2$ and $x = 3$ only. What are the zeroes of $p(x)$, and what is its likely degree? 2

23. A man tells his daughter, "Seven years ago I was 7 times as old as you were. Three years from now I shall be 3 times as old as you will be." Form the pair of linear equations (do not solve). 2

24. Represent the situation as a quadratic equation: The product of two consecutive positive integers is 72. 2

25. Which term of the AP $3, 8, 13, 18, \ldots$ is 63? 2

26. A ladder 17 m long reaches a window 15 m high. Find how far the foot of the ladder is from the wall. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Prove that $\sqrt{2}$ is irrational. 3

28. Find the zeroes of $x^2 - x - 6$ and verify the relationship between the zeroes and the coefficients. 3

29. On comparing the ratios, find whether the lines $5x - 4y = 8$ and $7x + 6y = 9$ intersect, are parallel or coincide. 3

30. Find the roots of $2x^2 - 7x + 3 = 0$ using the quadratic formula (if they exist). 3

31. In an AP, the 2nd term is 7 and the 5th term is 19. Find the AP and its 10th term. 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 4$ cm, $DB = 6$ cm and $AE = 12$ cm, find EC. 3

32. Find the value of k for which the points $(1,\ 2)$, $(3,\ 6)$ and $(5,\ k)$ are collinear. 3

OR

If $\tan\theta = \dfrac{12}{5}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Prove that $\sqrt{17}$ is irrational. 5

34. If α, β are the zeroes of $x^2 - 7x + 10$, form a quadratic polynomial whose zeroes are $\alpha + 1$ and $\beta + 1$. 5

35. A father is 8 times as old as his son. After 9 years the father will be 5 times as old as the son will be. Find their present ages. 5

36. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the journey. Find the original speed. 5

OR

How many terms of the AP $24, 22, 20, 18, \ldots$ must be taken so that the sum is 144? 5

37. Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. 5

OR

Find the point on the x-axis which is equidistant from $(7,\ 6)$ and $(-3,\ 4)$. 5

38. Prove that: $\dfrac{\sin\theta}{1+\cos\theta} + \dfrac{1+\cos\theta}{\sin\theta} = 2\csc\theta$. 5

OR

Two vertical poles of equal height stand on opposite sides of a road 80 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). 5

END OF QUESTION PAPER — **Total: 80 Marks**

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